

Sunny Bagal

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Profile

Backend engineer (NMIMS '26) who ships and measures: built a live RAG-powered link organizer (async ingestion, hybrid vector search) and a real-time collaborative whiteboard (append-only op-log, time-travel replay), then benchmarked both — 300× hydration speedup, 5× retrieval latency cut. Led 4 engineers as Technical Head intern at IntelCorp. Node.js · TypeScript · PostgreSQL · Redis.

Skills

Languages: Python, JavaScript, TypeScript

Backend: Node.js, Express, REST API design, WebSockets, BullMQ (message queues), JWT/OAuth 2.0, SSE, ORMs (Prisma, Drizzle)

Databases: PostgreSQL (SQL, pgvector), Redis (caching), MongoDB

Infra & Tools: Docker, Docker Compose, Linux, Git, CI/CD (GitHub Actions)

Frontend: React, Next.js, Tailwind CSS

Professional Experience

IntelCorp Software India Pvt Ltd, Technical Head (Intern)

01/2026 – 04/2026 | Navi Mumbai

- Led a 4-engineer team building a lead-generation platform; designed the core algorithm — recursive geospatial grid subdivision with ray-casting point-in-polygon filtering — to achieve exhaustive POI coverage where direct Maps queries truncate at ~120 results.
- Built the Node.js/Express backend end to end: Redis caching that eliminated repeat scrapes of previously-queried grids, a BullMQ pipeline for long-running geospatial jobs, and Stripe billing with webhook-driven plan entitlements.

Projects

Linea — Real-time collaborative whiteboard, TypeScript, Next.js, ws, Turborepo, PostgreSQL, Prisma, Canvas 2D

07/2026

- Built and deployed a real-time collaborative whiteboard (live: [linea.sunnybagal.com](#)); engineered the WebSocket sync layer from scratch (no socket.io): JWT-authenticated connections, room management, reconnect with exponential backoff, and a typed, Zod-validated message protocol.
- Designed state as a pure fold over an append-only, server-sequenced operation log — one reducer drives live sync, reload hydration, and a read-only time-travel slider; undo appends compensating ops, never rewriting history.
- Diagnosed an $O(n^2)$ hydration fold via benchmark scaling analysis and rewrote it as an $O(n)$ Map-fold with provably identical semantics — replay of 10k ops dropped 188ms → 0.6ms (~300×).
- Benchmarked the sync path at 800 acknowledged ops/sec through durable, atomically sequenced Postgres writes, with sub-30ms p95 broadcast fan-out (4 clients, local).

Recall — Link Organizer, React, TypeScript, Node.js, Express, PostgreSQL/pgvector, Redis, BullMQ, Claude API

05/2026

- Built and deployed a full-stack link organizer (live: [recall.sunnybagal.com](#)): BullMQ workers asynchronously fetch, summarize (Claude API), and embed each saved link into pgvector, keeping the API non-blocking end to end.
- Implemented hybrid retrieval — pgvector HNSW + keyword search fused with reciprocal rank fusion — on a custom retrieval stack (no LangChain) for full control of fusion and latency.
- Benchmarked and optimized retrieval: after an HNSW index initially showed no gain, per-arm profiling and EXPLAIN analysis isolated a Postgres cost-model misestimate; a scoped planner override cut p50 latency 46ms → 9ms (5×) over 10k vectors.
- Shipped an SSE-streamed RAG chat grounded in a user's saved links, with citation rendering.

NeuralCompress — Dual-Mode Image Compression System, PyTorch, CompressAI, DTCWT

01/2025 – 06/2025

- Designed a near-lossless compression pipeline combining Q-shift DTCWT with a reversible CNN (8 additive-coupling blocks, exact invertibility verified to $1e-5$), trained via uniform-noise quantization simulation — achieving ~100 dB PSNR for archival-integrity use cases.
- Built a matched-bitrate benchmarking harness comparing learned compression (CompressAI cheng2020-attn) against JPEG/WebP across quality sweeps; reproduced ~2× file-size reduction at matched PSNR (~33 dB @ 0.45 BPP vs JPEG's 0.98 BPP).
- Implemented a custom binary serialization format with zlib entropy coding, Sobel-saliency wavelet thresholding, and an artifact-removal CNN; served both modes via a Flask API with a live 4-way codec UI (demo: [huggingface.co/spaces/Sunny1110/image-compression](#)).

Education

B.Tech, Computer Engineering, Narsee Monjee Institute of Management Studies

2022 – 2026 | Navi Mumbai

CGPA: 7.38/10

Leadership & Activities

Co-Head, Creative Department — Ignite, NMIMS | 2023 – 2024

Led a 20+ member team for the college's flagship technical event; grew department social media reach 5.5×.

Volunteer, Jeevan Jyoti Ashalay (orphanage) — Nerul | 20-day on-site program

Coordinated event logistics and daily activities for resident children.

Certificates

AWS Academy Graduate — Cloud Architecting

Claude 101, AI Fluency, Building with the Claude API — Anthropic

Google Gen AI Study Jam — Google